

- 23 A uniform metal wire of length L and diameter d has spring constant k .

What is the Young modulus of the metal?

- A $\frac{\pi d^2}{4kL}$ B $\frac{\pi d^2}{kL}$ C $\frac{kL}{\pi d^2}$ D $\frac{4kL}{\pi d^2}$

- 24 An aircraft produces a sound at a frequency of 30.0 Hz.

The speed of sound in air is 330 m s^{-1} .

The aircraft is directly in front of the stationary observer and travels in a straight line towards or away from the observer.

The observer hears the sound from the aircraft at a frequency of 20.0 Hz.

What is the speed and direction of the aircraft?

	speed / m s^{-1}	direction
A	110	away from observer
B	110	towards observer
C	165	away from observer
D	165	towards observer

- 25 Which statement about longitudinal and transverse wave motion for a progressive wave is correct?

- A All transverse and longitudinal waves require a medium for propagation.
- B All transverse waves travel at the same speed, but longitudinal waves can travel at different speeds.
- C In both transverse waves and longitudinal waves, a particle with zero displacement has a maximum speed.
- D In a longitudinal wave there is a net movement of particles in the direction of travel of the wave, but there is no net movement of particles in a transverse wave.

- 26 The diagram shows a progressive transverse wave on a stretched string at one instant in time.

One point on the string is labelled X.

Which point on the string is 270° out of phase with X?

